

Shrutik Kupekar

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Summary

Software Developer with an **M.S. in Computer Science** at **Cal State Fullerton** with **2+ years of professional experience** building **full-stack, backend, and data-driven applications** using **React, Node.js, Python, SQL, and AWS**. Strong background in **system design, REST APIs, database optimization, and data validation**, with hands-on experience applying **LLM and RAG techniques** to production-style systems.

Technical Skills

Languages: JavaScript, TypeScript, Python, SQL, C, C++, Java

Web & Backend: React.js, Next.js, Node.js, Express.js, REST APIs, **System Design**, Django, Spring Boot

Data & AI: PostgreSQL (**pgvector**), MySQL, MongoDB, **Kafka**, Redis, **RAG**, LLM APIs (Claude, OpenAI), **Data Validation**, SQL Reporting, Power BI

Cloud/Tools: AWS (Lambda, API Gateway, DynamoDB, S3), Docker, Git, Linux, CI/CD, **Claude Code**

Core CS: Data Structures, Algorithms, Distributed Systems, Operating Systems, OOP

Experience

Software Developer — SAAA Consultants

Jun 2023 – Jul 2024

- Built and maintained **full-stack applications** using React, Node.js, Express.js, MySQL, and **REST APIs** for **20+ client projects**.
- Designed **backend APIs** and optimized database interactions to support reliable **data flow** between frontend, server-side logic, and relational databases.
- Delivered maintainable websites using React, Node.js, and MySQL, driving a **3x increase in client website traffic** through performance optimization and UI/UX improvements.
- Reduced page load times by **45%** through lazy loading, frontend optimization, debugging, and efficient **API design**.
- Led 2-3 client meetings per week, translating business requirements into technical solutions as the primary **technical point of contact**.

Software Developer & Tech Lead — Fineappl

May 2022 – Dec 2022

- Joined as the **first technical employee**, built the engineering function from the ground up, and helped grow the team to 5 members.
- Led **architecture and technical decisions** for a mental health application, working directly with the CEO on product and engineering strategy.
- Increased weekly active users by **70%** through UI/UX improvements, **offline-first data handling**, and background synchronization.
- **Mentored** junior team members and contributed to hiring and engineering practices.

Python Developer Intern — Cyberrace Infovision

Mar 2020 – Jun 2020

- Developed a **Python** analytics tool using Tkinter and MySQL to visualize, process, and analyze **structured datasets**.
- Automated **data validation** and processing workflows, improving accuracy and reducing manual effort.

Instructional Lab Assistant — California State University, Fullerton

Aug 2025 – Present

- Assist students in CPSC 120/121 with **C++**, debugging, OOP, memory management, control flow, and problem solving.

Volunteer Instructor — Learning by Locals NGO

Mar 2021 – May 2021

- Taught web development fundamentals, responsive design, clean coding practices, and hands-on frontend projects.

Projects

Serverless AI Chat Application — AWS, Lambda, API Gateway, DynamoDB, Next.js

- Built a serverless AI chat application using **AWS Lambda, API Gateway, DynamoDB, Next.js**, and OpenAI APIs, with no traditional always-running server.
- Designed **REST endpoints** for sending, retrieving, summarizing, and fact-checking messages with scalable serverless backend workflows.
- Integrated **AI-powered summarization and fact-checking** to help users review conversations, validate responses, and extract key insights.

AI News Intelligence Platform — FastAPI, Kafka, PostgreSQL (pgvector), Redis, Next.js

- Built a distributed news intelligence platform that ingests articles from RSS feeds/APIs, deduplicates and clusters them into "stories" using **semantic embeddings** and **pgvector**, and generates evolving "**what happened / why it matters / what's new**" summaries.
- Designed an **event-driven pipeline** using **Kafka** (raw to normalized to processed to embeddings to story-updates to alerts) with dedicated workers per stage, **Redis** caching, **S3** for raw storage, and **WebSocket**-based real-time story updates to the frontend.
- Implemented **RAG-based Q&A** grounded in cited source articles and stress-tested the pipeline from 10 to 500 articles/sec, hardening it with retries, dead-letter queues, and failover.

Education

California State University, Fullerton — **M.S. Computer Science**

Aug 2024 – May 2026

University of Mumbai — **B.E. Computer Engineering**

Aug 2018 – May 2022